HHS UAC PROGRAM

Predictive Forecasting of Care Load & Placement Demand

Complete Project Report • Data Analysis • Forecasting • Recommendations

A comprehensive, plain-language guide to the project — covering data, methods, findings, and 10 actionable recommendations for HHS decision-makers.

Program	HHS Unaccompanied Alien Children (UAC) Program
Objective	Predictive Forecasting of Care Load & Placement Demand
Data Source	Office of Refugee Resettlement (ORR) — Daily Time-Series
Models Used	SARIMA, Random Forest Regressor, Gradient Boosting Regressor
Tools	Python, Pandas, Scikit-learn, Statsmodels, Streamlit, Plotly
Output	Live Streamlit Dashboard + Forecast Reports for Stakeholders

Section 1: About the Data & Objectives

The dataset comes from the HHS UAC Program and tracks the daily movement of unaccompanied children through the U.S. border and federal care systems. Each row represents one reporting day and contains six key measurements:

Dataset Column	What It Means
Date	The reporting date for that row
Children Apprehended (CBP)	How many children were picked up at the border that day
Children in CBP Custody	Total children currently held by Border Patrol
Children Transferred to HHS	Children handed over from CBP to HHS care — the intake signal
Children in HHS Care	Total children currently in federal care — the main forecast target
Children Discharged from HHS	Children placed with sponsors — the exit/discharge signal

What This Project Set Out to Do:

- **Primary Goal 1:** Forecast how many children will be in HHS care over the next 7 to 30 days.
- **Primary Goal 2:** Predict the daily imbalance between intake (transfers) and exits (discharges).
- **Primary Goal 3:** Give early warnings before the care load reaches a dangerous level.
- **Secondary Goal:** Compare statistical models versus machine learning to find the most accurate approach.

Section 2: Importing Libraries & Loading Data

The project is built entirely in Python. The following tools were used:

- ✓ Pandas & NumPy — for loading, cleaning, and transforming the dataset.
- ✓ Matplotlib & Seaborn — for creating charts to explore the data visually.

- ✓ Scikit-learn — for building and evaluating the machine learning forecast models.
- ✓ Statsmodels — for the SARIMA statistical time-series model and seasonal decomposition.
- ✓ Streamlit & Plotly — for building the interactive dashboard for stakeholders.

The dataset is loaded as a CSV file, immediately inspected for structure and size, and the six key columns are confirmed. Each row represents one unique reporting date, giving us a clean daily time-series to work with.

🧹 Section 3: Data Cleaning & Preparation

Before any analysis, the raw data needed several corrections:

- ✓ Date column converted from text to a proper date format so the data can be sorted chronologically.
- ✓ Numeric columns contained commas (e.g., '2,484') — these were removed and values converted to numbers.
- ✓ Data sorted in chronological order from earliest to most recent date.
- ✓ Rows with missing values in the key target column (Children in HHS Care) were removed.
- ✓ Extra time features added: Year, Month, Week Number, and Day of Week — used as model inputs.
- ✓ A Net Flow column was created: Transfers minus Discharges. A positive number means the system is filling up; a negative number means it is emptying.

> After cleaning: the dataset was ready — a fully sorted, numeric, daily time-series with no gaps in the target variable.

📊 Section 4: What the Data Tells Us — Exploratory Data Analysis (EDA)

📈 **2023 Surge**	Children in HHS care peaked at 8,000–11,000 during mid-2023 — the highest levels in the dataset, driven by a border surge.
📉 **2025 Drop**	A sharp structural decline in early 2025 reduced care load to ~2,000–2,500. This was likely caused by policy enforcement changes — not a temporary dip.
🔄 **Daily Flow**	Average daily discharges (children placed with sponsors) stood at ~171/day — showing strong, stable placement capacity.
⚠️ **Pressure**	During 2023, transfers into HHS consistently exceeded discharges — creating sustained pressure that raised the care count. By 2025, discharges exceeded transfers, enabling the decline.
📌 **Old Plans Invalid**	Benchmarks built around 2023 peak levels (8,000–11,000 children) are outdated and no longer suitable for planning.
🔗 **Strong Autocorrelation**	Yesterday's care count is a powerful predictor of today's count. This confirms that lag-based features and time-series models are the right approach.

Section 5: Feature Engineering & Feature Selection

Raw data alone is not enough to build a good forecast. The following predictive features were carefully constructed:

Feature	What It Captures & Why It Matters
HHS_lag1 (Yesterday's count)	The strongest single predictor — today's count closely follows yesterday's.
HHS_lag7 (1 week ago count)	Captures weekly memory — useful for detecting patterns that repeat weekly.
HHS_lag14 (2 weeks ago count)	Captures medium-term memory — helps during gradual trend changes.
7-day Rolling Average	Smoothed recent average — reduces noise and shows short-term direction.
14-day Rolling Average	Broader smoothed trend — helps the model understand medium-term momentum.
7-day Rolling Std Dev	Measures recent variability — high values signal unstable or volatile periods.
Net Flow (Transfers−Discharges)	System pressure signal — positive means filling up, negative means emptying.
Day of Week, Month, Week No.	Calendar effects — captures weekly and seasonal operational rhythms.

All rolling features were calculated using a one-day shift to prevent the model from 'seeing the future' — ensuring the forecast is genuinely predictive, not artificially accurate.

Section 6: How the Model Was Tested — Train-Test Strategy

A strict time-based testing approach was used — the gold standard for forecasting models:

- **80% Training Data:** The model learned from data spanning 2020 through mid-2024 — covering normal operations and the 2023 surge.
- **20% Test Data:** The model was then tested on late 2024 through December 2025 — data it had never seen, including the 2025 structural decline.
- **No Random Sampling:** Data was never randomly shuffled. The model could only use past data to predict the future — exactly like real operations.
- **Walk-Forward Validation:** The model was re-evaluated across multiple time windows to confirm consistent performance — not just one lucky result.
- **Zero Data Leakage:** All lag features were computed using only information available at the time of prediction — no future data was used.

This is the same approach used in real-world forecasting systems. The model's reported accuracy is based on genuinely unseen data — meaning what you see is what you get in production.

Section 7: Seasonal Decomposition — Breaking Down the Care Load Signal

The care load data was mathematically split into 4 layers to understand what is driving the numbers at each level. This technique is called Seasonal Decomposition.

7.1 Observed Signal — The Raw Reality

- ✓ Shows the actual daily count of children in HHS care on every single day.
- ✓ Peak period: 8,000–11,000 children in 2023 to early 2024.
- ✓ A dramatic fall in early 2025 brought the count to approximately 2,000–2,500.
- ✓ This sudden change is a system-level regime shift — not a gradual seasonal change — and must be treated as the new baseline for all future planning.

7.2 Trend — The Long-Term Direction

- ✓ The trend rises steadily through 2023 and peaks in late 2023 or early 2024.
- ✓ A gradual decline follows through 2024, then a sharp structural break in early 2025.
- ✓ After the break, a mild recovery and stabilisation is visible — the system is not in freefall.
- ✓ Key insight: Capacity requirements have fundamentally reduced. Old budget and staffing plans built on 10,000-child benchmarks are now outdated and cost-inefficient.

7.3 Seasonality — The Weekly Rhythm

- ✓ A consistent weekly pattern repeats across the entire data series — intakes and discharges dip on weekends and rise on weekdays.
- ✓ This reflects normal operational schedules: caseworkers, courts, and sponsor placements are more active on weekdays.
- ✓ The weekly effect is small compared to the overall trend, so it should not drive major resource decisions alone.
- ✓ However, for daily roster planning, accounting for lower weekend activity can improve staffing efficiency.

7.4 Residuals — Shocks and Surprises

- ✓ Residuals are what remains after removing the trend and seasonal patterns — they represent unexpected events.
- ✓ During the early 2025 structural break, residuals spike sharply — confirming this was an external shock, not a predictable pattern.
- ✓ After the break, residuals return to small magnitudes — meaning the system is now behaving more predictably.
- ✓ Practical takeaway: The post-2025 period is MORE predictable than 2023–2024, making the current forecast more reliable than it would have been two years ago.

Section 8: Forecast Models & Performance — Which Model to Trust and Why

Model	Accuracy	MAE	MAPE	Verdict
☑ **Gradient Boosting**	~97.9%	~55	~2.1%	**Best — Use This**

Random Forest	~96.8%	~80	~3.2%	Good Backup
SARIMA (Statistical)	~85.9%	~320	~14.1%	Reports Only

> Plain Language: Gradient Boosting was wrong by only ~55 children per day on average — out of thousands. That is less than 2.1% error, making it reliable enough for daily operational planning.

Why SARIMA underperformed:

- ✓ SARIMA continued predicting a strong downward trend even as the actual data began recovering in 2025.
- ✓ It works best on stable, slowly changing data — but the UAC program experiences sudden policy-driven shifts.
- ✓ Despite lower accuracy, SARIMA remains useful for audit trails and stakeholder reports because its logic is easy to explain.

📏 Section 9: Key Performance Indicators (KPIs) — How We Measure Success

Five KPIs are tracked to measure both model performance and program health:

KPI	What It Measures	Why It Matters
Forecast Accuracy (%)	How close predictions are to actual counts	Determines if the model is trustworthy for planning
Surge Lead Time	Days of advance warning before a surge	Drives staffing and resource pre-positioning timelines
Capacity Breach Probability	Risk score for exceeding shelter capacity	Triggers early escalation and contingency plans
Forecast Stability Index	Consistency of predictions over time	Confirms the model does not produce wild swings
Net Flow Pressure	Transfers minus discharges (7-day average)	Real-time indicator of whether system load is rising or falling

💻 Section 10: Streamlit Dashboard — Live Analytics for Decision-Makers

A live, interactive web dashboard was built using Streamlit and Plotly to make all forecast outputs accessible to non-technical stakeholders. The dashboard includes:

- ✓ 5 KPI Cards at the top: Current care count, average discharges, net flow pressure, best model accuracy, 30-day forecast peak.
- ✓ Care Load Trends Tab: Historical daily and monthly trend chart, year-share donut chart, seasonal bar chart.
- ✓ Forecast & Models Tab: 30-day ahead forecast with confidence interval band, model comparison chart, MAE performance bar.
- ✓ Flow Analysis Tab: Weekly transfers vs discharges, net pressure chart (green/red bars), CBP pipeline chart.
- ✓ Recommendations Tab: 6 AI-generated operational recommendations with a full model scorecard table.

- ✓ Sidebar controls: Filter by year, switch forecast model, adjust forecast horizon (7–30 days).

> The dashboard auto-updates when new data is loaded and is designed for Monday morning weekly briefings — no technical knowledge required to interpret the outputs.

Section 11: 10 Recommendations for Stakeholders — Actionable Steps for the HHS UAC Program

Based on the full analysis — data exploration, seasonal decomposition, model training, and forecast evaluation — here are 10 concrete actions for the HHS UAC Program:

1 Deploy Gradient Boosting as the Official Forecast Tool

- ✓ This model achieved 97.9% accuracy on genuinely unseen data — the best of all three tested.
- ✓ It adapts to policy-driven changes better than statistical models like SARIMA.
- ✓ Use it as the primary engine powering the live dashboard and weekly forecast reports.
- ✓ Action: Replace any manual estimation with this automated forecast immediately.

2 Activate a 14-Day Surge Early Warning System

- ✓ The model can reliably predict care load 14 days ahead — enough time to prepare.
- ✓ When net flow pressure is positive for 3+ consecutive days, trigger a surge alert.
- ✓ This gives managers 2 full weeks to scale up staff, beds, and caseworker capacity.
- ✓ Action: Set Green / Yellow / Red thresholds in the dashboard for automatic alerts.

3 Update All Capacity Benchmarks to 2025 Levels

- ✓ The program now operates at 2,000–2,500 children — not the 2023 peak of 11,000.
- ✓ Maintaining old plans wastes budget on unused shelter space and unnecessary staffing.
- ✓ Right-size facilities to the current demand while keeping a defined surge reserve.
- ✓ Action: Conduct an immediate capacity review using 2025 data as the new baseline.

4 Schedule Monthly Automated Model Retraining

- ✓ Border conditions, policies, and migration patterns change — a static model degrades over time.
- ✓ Monthly retraining using the most recent 180 days keeps the forecast accurate.
- ✓ This requires no manual effort if automated through a simple scheduled pipeline.
- ✓ Action: Set up a monthly automated retraining job — takes less than 10 minutes to run.

5 Accelerate Sponsor Placements During Low-Intake Periods

- ✓ Average daily discharges of ~171 represent strong placement capacity that can be leveraged.
- ✓ During low-transfer periods, this capacity sits underutilised — a missed opportunity.
- ✓ Speeding up sponsor background checks during quiet periods builds a buffer for surges.

- ✓ Action: Pre-approve a pool of vetted sponsors to enable rapid placements on short notice.

6 Use Forecast Confidence Intervals for Budget Planning

- ✓ Every 30-day forecast includes an upper and lower uncertainty band — not just one number.
- ✓ Resource budgeting and staffing plans should be based on the upper bound for safety.
- ✓ This prevents under-preparation when actual counts come in higher than the average forecast.
- ✓ Action: Always present the confidence range in budget submissions and stakeholder reports.

7 Keep SARIMA for Audit Trails and Executive Briefings

- ✓ SARIMA is a traditional statistical model that regulators and auditors understand easily.
- ✓ While less accurate, its outputs can be explained in plain mathematical terms.
- ✓ Use it to cross-check Gradient Boosting results and for official compliance documentation.
- ✓ Action: Include SARIMA as a reference column in all auditable forecast records.

8 Publish a Weekly One-Page Dashboard Brief Every Monday

- ✓ Non-technical stakeholders need simple, visual updates — not raw numbers or model outputs.
- ✓ A one-page brief with traffic-light indicators (Green / Yellow / Red) is ideal.
- ✓ Include: current care count, 14-day forecast, discharge rate, and net flow pressure.
- ✓ Action: Automate this one-pager to generate from the Streamlit dashboard every Monday.

9 Add Policy Event Flags as Model Inputs

- ✓ The 2025 structural break surprised all models because it was a policy-driven external event.
- ✓ Future models should include binary flags for major enforcement actions or policy changes.
- ✓ This gives the model context to adjust predictions during regime-change periods.
- ✓ Action: Work with policy teams to maintain a dated log of enforcement events for model input.

10 Incorporate CBP Apprehension Data as a Leading Indicator

- ✓ CBP apprehension data precedes HHS transfers by several days — it is a natural early signal.
- ✓ Adding this could extend the reliable forecast horizon from 14 days to 21–30 days.
- ✓ Seasonal migration calendars and asylum filing trends can further improve early detection.
- ✓ Action: Establish a data-sharing agreement with CBP to feed apprehension counts into the model.